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ARTICLE

Self-Play Meta-Reinforcement Learning in Multi-Agent Games

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Abstract
Interactions in multi-agent systems are often framed through the tools of game theory; however, in real-world scenarios, the structure and parameters of the underlying game faced by agents are frequently unknown or non-stationary. This presents a critical challenge: agents must rapidly infer the nature of their environment and adapt their strategies accordingly, even in the presence of multiple other agents. Meta-reinforcement learning (meta-RL) has demonstrated the ability to facilitate fast adaptation in tasks such as multi-armed bandits, Markov decision processes, and visual navigation. In this paper, we extend the application of meta-RL to multi-agent games. By training agents via self-play meta-reinforcement learning on diverse classes of normal-form games, parameterized by their payoff matrices and sampled from a distribution, we develop algorithms that are not only sample-efficient and robust to changes, but also capable of strategic generalization across distinct game-theoretic structures. Although it remains limited to a theoretical proof of concept, our approach bridges the gap between classical game-theoretic modeling and modern meta-learning techniques, with promising implications for adaptive behavior in dynamic multi-agent environments.

Keywords Reinforcement-learning · Meta-learning · Game-theory

1 Introduction

Recent years have witnessed significant progress in the field of multi-agent reinforcement learning (MARL). A central theme in this research area is understanding the interaction dynamics among multiple agents, most often analyzed through the lens of game theory. Many foundational approaches such as Nash Q-learning [1] and its derivatives, multi-agent approaches in cooperative, competitive and coordination games have focused on repeated interactions within fixed game structures [2, 3], where the equilibrium conditions and strategic incentives remain relatively stable apart from possible stochastic fluctuations [4]. While these methods have yielded valuable insights, they inherit a common limitation of conventional reinforcement learning: the assumption that the training environment is identical to the test environment.

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This assumption is particularly problematic in real-world scenarios, where environments are often non-stationary and subject to continuous change. In multi-agent contexts, such changes may not only affect individual agent observations or rewards, but can also fundamentally alter the underlying game being played (for example shifts in market dynamics that modify the payoff landscape or strategic interdependencies among agents). Moreover, non-stationarity is already a core challenge in multi-agent settings, as agents continuously adapt to one another, altering the effective dynamics of the environment [5], and this endogenous non-stationarity is further compounded when the underlying game structure itself changes, making adaptation even more challenging. Robust MARL methods must therefore address both agent-induced and environment-induced sources of change.

While the broader challenge of distributional shift and environmental variability has been addressed in single-agent settings through meta-reinforcement learning, analogous progress in multi-agent systems remains limited.

Therefore, better MARL frameworks are needed that can facilitate rapid and robust adaptation to evolving game structures. Such frameworks should extend beyond fixed-game assumptions and develop mechanisms that allow agents to generalize across qualitatively different strategic settings.

Meta-learning, or "learning to learn," focuses on training models that can quickly adapt to new tasks by leveraging experience from a distribution of related tasks [6]. Unlike traditional machine learning approaches that optimize performance on a single fixed task, meta-learning frameworks aim to discover learning strategies that generalize across a family of tasks. This enables agents to adapt efficiently when faced with new, unseen situations, making meta-learning particularly appealing in environments where flexibility, fast adaptation, and robustness are critical.

The present study introduces a novel multi-agent meta-learning framework that adapts single-agent meta-reinforcement learning techniques to normal-form games in the multi-agent setting. We generate distributions of these games by parameterizing their payoff matrices and focusing on key strategic properties, such as cooperation (coordination games) and conflict (zero-sum games). We demonstrate that this approach enables agents to rapidly adapt and learn robust joint policies across a diverse range of strategic environments, thus emphasizing its effectiveness in handling variability and complexity in multi-agent interactions.

However, it is important to mention that although the importance of strategic diversity and adaptability to environmental drifts is prominent in real-world problems, experiments at this level of analysis are conducted in generalized theoretical scenarios such as the normal-form games this paper uses. This way of conducting experiments has the disadvantage of not being applied in any particular real-world setting, and the advantage of generalizability across many. Thus we design and conduct several types of experiments in order to empirically investigate the following research questions:

- **Q1**: Can independently trained self-play RL^2 co-adapt and converge to Nash equilibria in completely decentralized multi-agent settings in distributions of normal-form games?
- **Q2**: How well does RL^2 scale to game-theoretic environments with varying action spaces?
- **Q3**: Under what conditions is independent self-play more effective than fictitious self-play in reaching equilibrium?

The rest of this paper is organized as follows: Section 2 introduces the necessary theoretical background of our work including game theory, definitions and solution concepts, respectively reinforcement learning and meta learning. Then, Section 3 discusses related work,

highlighting similarities and differences between existing literature and our work. Section 4 elaborates on the design of our system, describing agents' architecture, the families of games we consider, along with the hyperparameters used. Further, Section 5 discusses the results obtained, and finally in Section 6 conclusions are drawn.

2 Background

2.1 Game theory

Game theory provides a powerful framework for modeling strategic interactions between decision-makers, making it a natural complement to reinforcement learning in multi-agent settings. By formalizing notions such as optimality, best response, and equilibrium, game theory helps us understand how rational agents might behave when their outcomes depend on others' actions. Formally, a game is defined by a tuple $G = (N, \{A_i\}_{i \in N}, \{u_i\}_{i \in N})$ where N is a finite set of players, A_i is the set of actions available to player i, and $u_i : A_1 \times ... \times A_n \to \mathbb{R}$ maps the joint actions to real-valued payoffs for player i. A strategy for player i is a probability distribution over A_i, defined as $\sigma_i \in \Delta(A_i)$, where $\Delta(A_i)$ denotes the set of all possible distributions over A_i. A strategy is called pure if only one action has probability 1 and all the rest are zero, and mixed otherwise. A joint strategy profile $\sigma = (\sigma_1, ..., \sigma_n)$ is a tuple of strategies for all players.

One of the most central solution concepts is the Nash equilibrium, where no player has an incentive to unilaterally deviate from their strategy. Formally, a joint strategy profile $\sigma = (\sigma_1, ..., \sigma_n)$ is a Nash equilibrium if, for each agent i, it satisfies that given that all other agents play their equilibrium strategies, no agent can improve its expected payoff by changing its own policy, that is, $u(\sigma_i, \sigma_{-i}) \geq (\sigma_i', \sigma_{-i}), \forall i \in N, \forall \sigma_i' \in \Delta(A_i)$, where σ_{-i} denotes the set of all strategies from σ except σ_i.

In the context of learning, equilibrium concepts serve both as benchmarks and guiding principles to evaluate and interpret agent behavior. While games can be represented in multiple ways (such as extensive form, graphical form, or state-based formulations) we focus on normal-form games, where each player selects a single action per interaction from a fixed action set represented by so called payoff matrices. This representation is not only compact and expressive for simultaneous-move games, but also lends itself well to parameterization and efficient sampling, which are essential for large-scale experimentation and meta-learning.

2.2 Meta-Reinforcement Learning

Meta-learning, or "learning to learn," is a collective term denoting a class of methods that aim to acquire inductive biases or learning strategies across a distribution of tasks, enabling rapid adaptation to new tasks with minimal data [7]. Formally, a meta-learner optimizes its parameters not only to perform well on a single task, but to generalize across tasks drawn from a task distribution. A key feature of meta-learning systems is their ability to use information from previous experiences–such as gradients, predictions, or intermediate outputs–to guide future learning [8]. In meta-reinforcement learning, this principle is applied to reinforcement learning environments: agents are trained across multiple MDPs or environments such that they can adapt quickly to new ones [9, 10].

In RL, the goal is to find a policy function π in a Markov Decision Process (MDP) that maximizes reward [11]. Formally, an MDP is defined as a tuple (S, A, P, R, γ), where S

is the set of possible states, A is the set of possible actions, $P : S \times A \times S \rightarrow [0, 1]$ encapsulates the transition probabilities to describe the dynamics of the environment, $R : S \times A \rightarrow \mathbb{R}$ is a reward function and γ is called the discount factor that controls the short- or farsightedness of the agent. Complementary to γ, some formulations also add a horizon T to this definition, that denotes a possible maximum length of episodes. While in simple RL, the policy $\pi : (s_t) \rightarrow \Delta(A)$ maps states to distributions of actions, in meta-RL, the policy $\pi : (a_{t-1}, r_{t-1}, s_t) \rightarrow \Delta(A)$ maps previous actions, previous rewards and current states to such distributions. The core idea behind this design is feeding a history of transitions into the agent so that it can internalize the dynamics between states better and faster.

Multi-Agent Reinforcement Learning (MARL) [12] extends the idea of MDP to the so-called Markov games. A Markov game is defined by a tuple $(N, T, S, \{A_i\}_{i \in N}, \{r_i\}_{i \in N}, P)$, where N is the finite set of agents, S is a finite set of states, A_i denotes the actions available to agent i. $r_i : \{T\} \times S \times A_1 \times ... \times A_n \rightarrow \mathbb{R}$ is the reward function for agent i while $P : \{T\} \times S \times A_1 \times ... \times A_n$ is the transition kernel. Further, a policy of agent i is defined as $\pi_i : \{T\} \times S \rightarrow \Delta(A_i)$, a joint action $a = (a_1, ..., a_n)$ and a joint policy is $\pi = (\pi_1, ..., \pi_n)$. Similarly to Section 2.1, we note by $-i$ all agents except i. Then, the state value functions can be written as:

$$V^{\pi}_{t,i}(s) = \mathbb{E}_{\pi}[\sum_{t'=t}^{T} r_{t',i}(s_{t'}, a_{t'})|s_t = s] \tag{1}$$

Analogously, the Q function is defined as:

$$Q^{\pi}_{t,i}(s, a) = \mathbb{E}_{\pi}[\sum_{t'=t}^{T} r_{t',i}(s_{t'}, a_{t'})|s_t = s, a_t = a] \tag{2}$$

Then, a policy π^*_i is a best response to π_{-i} if $V^{\pi^*_i, \pi_{-i}}_{1,i}(s1)$ maximizes $V^{\pi_i, \pi_{-i}}_{1,i}(s_1)$ across all possible π_i. Then, a joint policy $\pi = (\pi_i, \pi_{-i})$ is a *Nash equilibrium* if π_i is a best response to $\pi_{-i}, \forall i \in N$.

In multi-agent meta-learning settings, agents are trained on a set of Markov games drawn from a distribution, generally sharing the same state and action spaces but with different reward functions and transition kernels [13, 14].

Typically, meta-reinforcement learning settings are structured into two nested loops, an inner and an outer loop. At the beginning of each meta-episode (the outer loop, also called *trial*), a new environment is drawn from a predefined task distribution, collecting multiple episodes but maintaining hidden state (usually through a recurrent neural network or Gated Recurrent Units [15]). The hidden state is generally reset between tasks to prevent information leakage across tasks. After a fixed number of episodes, the collected trajectories are used to compute a meta-loss, which is used in the outer loop to update the agent's parameters, thus improving its general ability to learn and adapt in future unseen environments.

RL^2 [16] is a meta-reinforcement learning framework that aims to produce agents capable of rapidly adapting to new environments by encoding past interactions into a recurrent policy. Instead of hand-crafting fast adaptation mechanisms, RL^2 treats the learning algorithm itself as part of the policy, trained end-to-end via reinforcement learning, and considering the outer loop as a RL problem too.

This approach has been effective in domains such as bandits, MDPs, and navigation, where prior experience helps guide exploration and strategy in novel tasks. While most prior work focuses on single-agent settings, extending RL^2 to multi-agent scenarios introduces new challenges and opportunities. In particular, independently but simultaneously training

multiple RL^2 agents (each adapting based on local experience without coordination) mirrors many real-world situations where agents interact strategically without centralized control.

Additionally, fictitious self-play offers a complementary perspective by modeling opponents as stationary policies that evolve slowly over time, providing a stable yet nontrivial learning signal that encourages strategic adaptation across repeated interactions. In practice, for two-player games this often means training only one agent against fixed past versions of itself.

Studying such interactions in game-theoretic environments is especially valuable, as these settings offer well-defined solution concepts, strong theoretical foundations, and clear metrics for measuring adaptation and robustness. Zero-sum games, in particular, serve as a rigorous and high-signal testbed for probing agent behavior, strategic adaptation, and stability under learning dynamics.

3 Related work

Understanding strategic interactions in multi-agent settings through the lens of game-theoretic solution concepts has long been a research focus [1, 17], and interest has grown significantly following successes in games such as Go [18] and StarCraft [19], making MARL a popular research area [20, 21].

In the domain of single-agent meta-reinforcement learning, the core idea is the one of memory-based agents [22]. In this field, two foundational approaches stand out: RL^2 [16] and Model-Agnostic Meta-Learning (MAML) [23]. RL^2 treats the meta-learning process itself as a reinforcement learning problem, training a recurrent policy to adapt through experience. In contrast, MAML is a gradient-based method that optimizes for an initialization that can adapt quickly to new tasks via a few gradient steps. MAML has since inspired several extensions in game-theoretic MARL, such as [24].

On the more explicitly game-theoretic side, key subfields are regret minimization learning [25, 26] and finding Nash equilibria with deep neural networks. In [27] an investigation of the learnability of Nash equilibria is conducted, while in [28], a network that rapidly and deterministically approximates equilibria in normal-form games is proposed. Further, [29] uses generative adversarial methods to learn equilibria, while [30] proposes a method that transforms normal-form games into action embeddings, and evaulate their method on several game-theoretic benchmarks to highlight versatility and adaptiveness. Meta-learning based Nash equilibrium approximators have also been proposed in combination with counterfactual regret minimization [31]. Although this method does not guarantee convergence to Nash equilibria, empirical results in general-sum normal-form games demonstrate convergence to coarse correlated equilibria.

The most closely related work to ours is that of [13], which also addresses similar classes of games, considering zero-sum games and potential games, while measuring the Nash-gap to evaluate performance related to achieving equilibria. The authors further investigate theoretical conditions under which meta-learning converges to equilibrium strategies. The proposed algorithm adopts a MAML-based approach, and their evaluation is limited to a fixed set of ten games and does not account for variability in action space size. Moreover, while the variety in tasks is only achieved by perturbing one game, our experiments are much more comprehensive in this respect since we fully randomize payoff matrices and introduce minimal assumptions otherwise. Additionally, our experiment considers a variety across action space sizes, and a comparison between two different meta-training schemes

too. Nonetheless, the theoretical results in [13] are still a significant milestone in the field of meta-RL.

Finally, [32] introduces a meta-game evaluation framework for MARL systems, benchmarking various algorithms across a family of negotiation games. While their focus is not on meta-learning per se, the meta-game structure that they also define as a distribution over games is conceptually aligned with our goals, focusing on measuring adaptability and robustness.

4 Design

This section is meant to elaborate on a high-level overview of our meta-learning framework, followed by a presentation of the proposed families of games and their parameterizations.

4.1 Architecture

Reinforcement learning in competitive settings presents significant challenges due to non-stationarity and the lack of convergence guarantees for most single-agent algorithms. In two-player zero-sum games, traditional learning dynamics often fail to converge to equilibrium strategies unless strong assumptions are made about the game's structure or the learning algorithm's update rules. Moreover, in realistic environments, the underlying game dynamics may change over time or across scenarios, demanding agents that can adapt quickly and reliably to unseen variations not only in opponent strategies but the game itself as well.

To address these challenges, we adopt a modified version of RL^2 [16] as our meta-learning framework, training two independent agents to adapt to a distribution over multiple types of games, focusing on two-player games only. By framing the learning process itself as a reinforcement learning problem, RL^2 enables agents to internalize a learning algorithm within a recurrent neural network, conditioned on past experiences. This structure implicitly encodes exploration, adaptation, and exploitation strategies through memory, rather than through hand-crafted updates. The resulting agents are memory-augmented meta-learners that do not merely optimize for performance in a fixed game, but instead learn how to learn across an entire environment class. This approach aligns naturally with the demands of multi-agent competition, where robustness and adaptability are not just desirable but essential. We hypothesize that via independent RL^2 learners, more robust and adaptable agents can be developed. Figure 1 offers a high level depiction of how the meta-RL training scheme with games is conceived.

4.2 Zero-sum games

Zero-sum games represent competitive interactions in which one player's gain is exactly the other player's loss. In two player games, this implies that given a payoff matrix A for player 1, the corresponding matrix for player 2 is $-A$.

This structural symmetry eliminates trivial cooperative solutions and inherently drives the need for strategic reasoning and continuous adaptation. As a result, robustness and adaptability become central to learning in such environments. Moreover, these games have well-defined and well-understood solution concepts, especially related to Nash equilibria, making zero-sum games a high-signal environment with mathematically grounded ways of interpretation and measurement of performance.

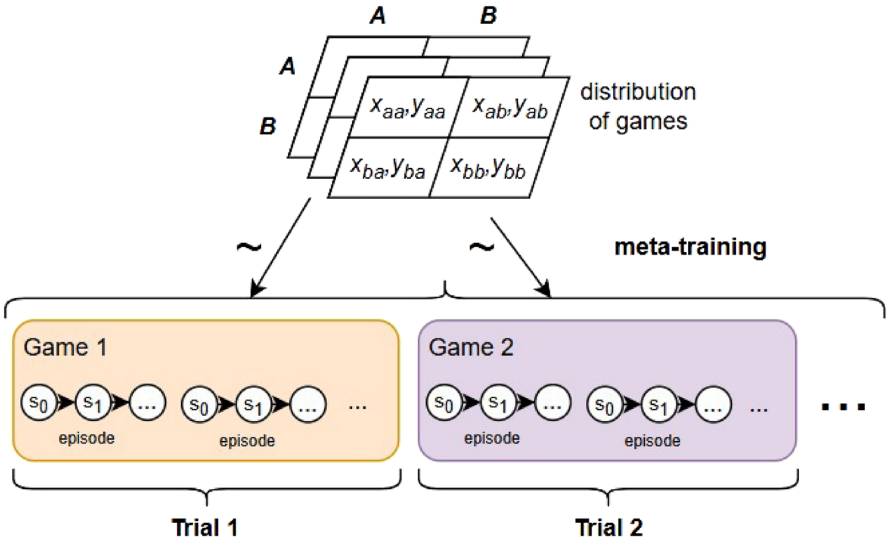


Fig. 1 High level overview of multi-agent RL^2 with randomly sampled games

Table 1 Payoff matrix for a 3x3 zero-sum game from M_2^3

	a	b	c
a	(1.45 -1.45)	(-0.41 0.41)	(-1.83 1.83)
b	(-1.09 1.09)	(-0.63 0.63)	(1.49 -1.49)
c	(-0.76 0.76)	(0.17 -0.17)	(1.44 -1.44)

In our setting, we aim to train agents capable of generalizing across a broad spectrum of zero-sum games. To this end, we define a distribution over games with minimal structural assumptions. Specifically, we consider the family $M_x^m = \{(A, -A) | A \in \mathbb{R}^{m \times m} A_{ij} \sim U(-x, x)\}$ where $U(-x, x)$ denotes the continuous uniform distribution between x and -x. Sampling from a continuous distribution ensures that, with probability one, each game in the family admits a unique Nash equilibrium in mixed strategies [33, 34]. Table 1 shows a sample payoff matrix drawn from the distribution described above. The unique mixed-strategy Nash equilibrium in this example is $\sigma_{row} = (\frac{1}{3}, 0, \frac{2}{3})$ and $\sigma_{column} = (\frac{1}{5}, \frac{4}{5}, 0)$ for the row and column players respectively, where each array denotes the probability of playing actions a, b, and c in this order.

4.3 Cooperative games

To study coordination under partial cooperation and ambiguity, we introduce a custom cooperative game with a carefully controlled payoff structure. The game is constructed by first sampling each entry in the joint payoff matrix independently from a uniform interval $(-x + \epsilon, x - \epsilon)$, where x is a predefined value radius and $\epsilon > 0$ ensures that the baseline payoffs remain suboptimal. We then select a random joint action (i.e., a matrix position) and assign both players the maximum payoff x for that action, creating a single, sharply optimal equilibrium embedded within a noisy reward landscape. We denote this family of games as $M_{x,\epsilon}^m$. Moreover, the distribution over games itself, and more specifically, the random

Table 2 Sample cooperative game drawn from $M^3_{2,0.5}$

x	a	b	c
a	(0.43 -1.36)	(-1.33 0.78)	(-0.85 1.09)
b	(0.9 -0.47)	(-0.50 1.22)	**(2.00 2.00)**
c	(-0.03 1.00)	(1.22 -0.12)	(0.55 -0.16)

placement of the optimal payoff entry serves as a natural source of adaptive pressure for the agents.

This setup resembles a noisy potential game, but with a stronger incentive structure: although multiple Nash equilibria typically exist, only one equilibrium yields substantially higher rewards, making it a natural coordination target. This construction is particularly advantageous for interpretability: in general, the existence of multiple equilibria complicates the application of equilibrium-based regret metrics. However, by combining these metrics with achieved reward statistics, we gain a richer picture of agent performance, revealing not only whether agents converge to equilibria, but also whether they successfully identify the most socially beneficial one. Table 2 shows a sample cooperative game. In this example, the most beneficial pure strategy Nash equilibrium is playing c for the column player and playing b for the row player.

4.4 Training

We build upon the original RL^2 framework, with a key modification: instead of using TRPO [35] for policy optimization, we adopt Proximal Policy Optimization (PPO) [36], which has since demonstrated to offer better empirical stability and efficiency, particularly in recurrent architectures. Algorithm 1 outlines our training scheme in detail. At each trial, a new game is sampled from a game family $\mathcal{M}^{(m)}$ characterized by a reward range x and m available actions. The environment is then interacted with for K episodes. Policy optimization is performed jointly for both agents at the end of each trial, ensuring balanced and simultaneous learning. The reward $r_{i,t}$ for agent i is computed as the payoff received from the game after the joint action $a_t = (a_{0,t}, a_{1,t})$ is executed in the game described by matrices (A, B).

This setup leads to a fully decentralized training and execution regime, where both actors and critics are independent and receive no shared information, which in general is an arrangement known to pose greater challenges for convergence. In experiments using fictitious self-play, the training changes such that the second agent is not updated. Instead, at the beginning of each trial P, the second agent's policy is set to θ_{P-1}, the snapshot of the first agent's policy from the previous trial, and analogously θ_P is saved from the first player. As such, fictitious self-play in our implementation only incorporates updates to the opponent's policy on a per-trial basis. This update frequency of the opponent's policy departs from traditional fictitious self-play [37], since traditionally there is no learning outer-loop, and our fictitious self-play respects the original in the sense that agents play against previous versions of themselves.

Since the normal-form games we consider are simultaneous-move games, the labels "first" and "second" player are used solely for clarity of exposition. As a result, there is no inherent asymmetry between the agents, and there is no need to alternate which agent is being trained during fictitious self-play.

Algorithm 1 RL^2 Training for Two-Agent Normal Form Games

Require: $\mathcal{M}^{(m)}$ - game distribution
N - number of training iterations
K - episodes per game
T - episode length
Ensure: Meta-trained policy networks π_{θ_1} and π_{θ_2}
Init recurrent policy networks $\pi_{\theta_1}, \pi_{\theta_2}$ with params θ_1, θ_2
for each training iteration **do**
  Sample game $(A, B) \sim \mathcal{M}_x^{(m)}$
  Initialize hidden states h_1, h_2
  for $k = 1$ to K **do**
    Initialize environment state s_0
    for $t = 0$ to $T - 1$ **do**
      $a_{1,t} \sim \pi_{\theta_1}(a \mid s_t, h_1)$
      $a_{2,t} \sim \pi_{\theta_2}(a \mid s_t, h_2)$
      Compute payoffs: $(r_{1,t}, r_{2,t}) \leftarrow (A_{a_{1,t},a_{2,t}}, B_{a_{1,t},a_{2,t}})$
      Update h_1, h_2 with $(s_t, a_{1,t}, r_{1,t}), (s_t, a_{2,t}, r_{2,t})$
      update s_{t+1}
    end for
  end for
  Compute policy gradients from accumulated trajectories
  Update θ_1 and θ_2 via PPO
end for

4.5 Evaluation

To evaluate agent performance in the meta-reinforcement learning setting, we report two complementary metrics: the average Nash regret and the average episode reward, both computed per meta-episode. The sum of Nash-regrets, also known as the Nikaido-Isoda [38], NashConv [39] and in other forms, average-deviation incentive [40] is a widely used metric in game-theoretic research similar to this study. While the reward provides a task-specific signal reflecting the agent's behavior, it is not necessarily aligned with equilibrium optimality in multi-agent scenarios and is thus used only for explanatory and interpretive goals. Our primary metric of interest is the *Nash regret* which for a game $G = (N, \{A_i\}_{i \in N}, \{u_i\}_{i \in N})$ and strategy profile σ is defined as

$$Regret_i(\sigma) = max_{\sigma_i'}(u_i(\sigma_i', \sigma_{-i})) - u_i(\sigma) \tag{3}$$

which measures how much more reward agent i could have gotten versus its current strategy. We estimate joint strategy profile σ from player histories, that is, at the end of each episode, the empirical action frequency is estimated from past actions within that episode. Then, histories are reset after each episode and contrary to the hidden state, they are not persisted per meta-episode. The value $\sum_{i \in N} Regret_i(\sigma)$ then measures how much the entire set of players deviates from best responses; thus, when this sum is zero, a Nash equilibrium is achieved. We then average these per-episode regret values at the meta-episode level and report the outer loop averaged meta-episode rewards. We empirically show that our agents tend to minimize this value albeit only being fed the actual game rewards and have no explicit knowledge of this metric. We refer to those experiments where one agent is trained against past versions of itself as fictitious meta-self-play experiments, while those where two entirely independent agents are co-trained, as independent co-training.

Table 3 Hyperparameters used for PPO

Param name	Param value
actor learning rate	0.0001
critic learning rate	0.0003
clip range	0.1
entropy coefficient	0.01
gae-lambda	0.3

4.6 Detailed experiment setup

To address our research questions, we design experiments involving meta-training of RL^2 agents using two distinct learning schemes: *independent co-training*, where agents are trained concurrently without explicit modeling of others, and *fictitious self-play*, where agents iteratively adapt to a fixed version of the opponent. We evaluate these methods across games with varying action space sizes: we use 2, 3, 4, 5, and 6 actions per player. We avoid larger action spaces due to the increasing interpretability challenges posed by high-dimensional tabular normal-form games.

Two game types are considered: zero-sum games, which highlight competitive dynamics, and a custom-designed coordination game, which emphasizes cooperative behavior and joint optimization. To contextualize performance, we compare against three baselines: a random policy, used as an absolute lower-bound reference; a value-based feedforward model, which illustrates the limitations of conventional RL approaches in generalizing across diverse strategic settings; and a recurrent benchmark model, which offers a fair and competitive comparison for the RL^2 architecture. For a theoretical upper bound, we use the regrets accumulated after running extra-gradient [?] limited to ten iterations per game that we label EG10. Running extra-gradient for more than $\sim$100 steps usually solves these games completely and generally accumulates no regret.

A more appropriate meta-learning baseline model would be MAML adapted to RL, however as far as our experiments show, independent MAML agents are incredibly difficult to tune [41] and tend to diverge in the fully decentralized co-training setting. Despite being one of the most successful meta-learning method, convergence is difficult to achieve and results tend to be difficult to reproduce [42] and thus the integration of MAML into decentralized-training decentralized-execution RL setups remains an open challenge.

For the game distribution parameters, we use a payoff range of $(-2, 2)$ for zero-sum games, and $\epsilon = \frac{1}{2}$ with a payoff range of $(-2, 2)$ for cooperative games.

Each experiment involves 100 meta-training epochs, with 1,000 meta episodes per evaluation, and 50 interaction steps per game.

GRU blocks are used to store the hidden state with 256 hidden units, and PPO for optimization. Both agents (in the case of independent training) are symmetrical in the underlying model structure. Table 3 summarizes the hyperparameters our experiments use in the outer loop optimization of the RL^2 agents.

Each experiment was averaged over 10 independent runs, and the reported results reflect the mean Nash regret per meta-episode and mean reward per meta-episode, further averaged over the final 5 meta-episodes of training to smooth out variability and capture stable performance.

5 Results

This section presents the empirical findings from our experimental study. In addition to evaluating the proposed approaches including MAML, fictitious self-play and independent co-training with RL^2, we benchmark performance under the same conditions (but without an outer loop optimization) as our RL^2 agents against three baselines: a random policy, Deep Q-Network (DQN) agents [43], and recurrent PPO agents [44], providing a comprehensive comparison across learning paradigms. Moreover, a theoretically guaranteed oracle baseline is also considered, consisting of the regret values achieved by an extra-gradient algorithm after 10 iterations.

5.1 MAML-based methods

To facilitate a comparison of meta-RL methods, fully independent MAML-based agents were co-trained following a protocol analogous to the RL^2 methods. Similarly to RL^2, a number of 100 outer loops were considered, evaluating 1,000 games with 50 interaction steps each. For each agent, a single gradient step was utilized, employing a trust region of 0.01, 10 conjugate gradient steps with damping coefficients between 10^{-5} and 10^{-3}, and a line search with a backtrack ratio of 0.8 (maximum 15 steps). Network architectures with hidden sizes of 32x32 and 64x64 were evaluated, along with GAE-λ values ranging from 0 to 1, discount factors (γ) between 0.8 and 1, and learning rates from 10^{-4} to 0.1. However, these MAML agents were observed to be highly unstable, a result attributed to the difficulties of the outer TRPO optimization during the co-adaptation step and no conclusive levels of convergence have been observed.

5.2 Zero-sum games

In our experiments, we observe that independent co-training of two RL^2 agents is generally effective at minimizing Nash regret, particularly for games with small to moderate action spaces. While the agents do not achieve significantly higher cumulative rewards as the action space increases, their reward trajectories remain relatively stable, suggesting a consistent adaptive pressure driven by the distribution of games. Moreover, the agents' individual meta-episode regrets are also well aligned, which further strengthens the former argument.

This pattern changes notably under fictitious self-play. Here, the training agent exhibits steadily increasing cumulative rewards, implying that it primarily learns to exploit the fixed opponent from previous meta-episodes. Although the resulting strategies still approximate equilibrium behavior to some extent, as indicated by regret values, the fictitious self-play agents are consistently outperformed by those trained via independent co-training. This outcome strongly suggests that updating the fixed agent more frequently than once per meta-episode could mitigate the imbalance in strategic incentives, reduce overfitting to one's past self, and ultimately lead to more robust coordination and generalization across games. Table 4 summarizes these findings in zero-sum games. The values of m denote the numbers of actions considered. Fict. RL^2 and Ind. RL^2 stand for fictitious self-play, and independent co-training experiments respectively.

It is important to note that naive DQN agents often perform slightly worse than even random policies. This counterintuitive result stems from the fact that value-based methods lack the ability to retain shared structure across tasks and must effectively re-learn from scratch for each newly sampled game. The constant cycle of forgetting and re-adaptation

Table 4 Nash-regret values for zero-sum games

m	Random	DQN	PPO	Fict. RL^2	Ind. RL^2	EG10
m=2	0.67 ± 0.016	0.68 ± 0.030	0.45 ± 0.024	0.28 ± 0.009	0.23 ± 0.014	0.24 ± 0.113
m=3	0.99 ± 0.014	1.03 ± 0.020	0.85 ± 0.012	0.59 ± 0.010	0.54 ± 0.009	0.34 ± 0.116
m=4	1.19 ± 0.020	1.26 ± 0.042	1.17 ± 0.008	0.79 ± 0.007	0.79 ± 0.014	0.37 ± 0.117
m=5	1.35 ± 0.016	1.37 ± 0.030	1.37 ± 0.027	1.01 ± 0.016	1.04 ± 0.006	0.37 ± 0.116
m=6	1.45 ± 0.022	1.48 ± 0.024	1.47 ± 0.018	1.28 ± 0.010	1.27 ± 0.010	0.38 ± 0.107

imposes a significant learning burden in the meta-learning setting, leading to degraded overall performance despite their theoretical capacity for convergence within individual games.

Analogously, even recurrent PPO struggles to outperform the random baseline by a significant margin. While having memory provides some advantage by enabling short-term pattern recognition, the absence of an explicit outer loop optimization means the agent still suffers from the same core issue: it must repeatedly forget and re-learn across episodes, as it lacks a mechanism to accumulate and generalize knowledge across tasks. Given that our environments are fully randomized at each meta-episode, this inability to meta-adapt makes it difficult for the agents to exploit recurring structural patterns. The effects of this missing outer loop are clearly observable in the results. Nevertheless, we include these baselines to underscore the impact and value of meta-RL approaches.

It is also worth noting that as the action space increases, the performance gains over random agents tend to diminish and depart significantly from theoretical bounds. This trend suggests that larger and more complex action spaces may require correspondingly more expressive models to effectively capture the underlying strategic structure. In our experiments, we deliberately used a fixed architecture with the same number of hidden units and parameters across all settings; the observed degradation in performance therefore highlights a potential limitation in representational capacity, rather than learning dynamics alone. Moreover, even a modest increase in the individual action space has a compounding effect: since the joint action space grows combinatorially with the number of actions per agent, the learning and generalization challenges escalate rapidly, which motivates our modest variety of action space sizes.

Figure 2 illustrates the evolution of Nash regret and accumulated reward throughout training. The top row presents the results of independent co-training, while the bottom row corresponds to fictitious self-play. On the left, we report the mean Nash regret over meta-episodes for each outer loop, and on the right, the mean meta-episode reward averaged for each outer loop. In the top-left plot, we report the individual rewards of both agents rather than their average, as averaging would be uninformative in the zero-sum setting. The x axis denotes the evolution of these metrics. Both regret minimization and opponent exploitation in fictitious self-play become increasingly slower as the size of the action space grows.

5.3 Coordination games

In the cooperative game setting, the performance advantage of independent co-training over fictitious self-play becomes even more pronounced. This can be attributed to the fact that effective cooperation requires both agents to adapt simultaneously, gradually converging toward mutually beneficial strategies. Independent co-training facilitates this by allowing



Fig. 2 The evolution of Nash regrets (left) and cumulative rewards (right) in zero-sum games. The x axis shows the evolution of time (one value reported per outer loop) while the y values represent the actual regret- and reward values respectively

both policies to continuously co-evolve, promoting synergy and coordination. In contrast, fictitious self-play constrains one agent to optimize against a fixed partner for the entire trial, which can hinder the emergence of coordinated behavior, particularly when the fixed policy is suboptimal or outdated.

As a result, the rigid dynamics of fictitious self-play can limit the discovery of high-reward joint strategies, because only the actively learning agent explores continually, although there is a need for both to exploit the higher reward yielding payoff matrix entry. This makes independent co-training a more effective approach for meta-learning in cooperative environments. Indeed, the outer loop optimization plays a central role in enabling independent co-training to effectively handle two distinct sources of non-stationarity. First, it allows agents to adapt to changing opponents, learning strategies that are robust to the evolving behaviors of other agents encountered across meta-episodes. Second, it supports generalization across diverse, randomly sampled environments, enabling the agent to accumulate inductive biases that help it identify and exploit structure more efficiently. This dual adaptability to both the strategic landscape of the current task and to the task distribution, is what sets meta-trained agents apart. In contrast, approaches lacking an outer loop, such as recurrent PPO or naive value-based methods such as Q-Learning [1], are inherently limited in their ability to retain and transfer knowledge, forcing them to restart learning anew for every new task. However, these results also suggest that fictitious meta self-play where opponent updates are done within the trials too might be promising to explore, although the non-stationarity that comes from both agent-variety and the inherent one from the meta-learning setup, remains challenging.

Table 5 Nash-regret values for coordination games

m	Random	DQN	PPO	Fict. RL^2	Ind. RL^2	EG10
m=2	0.74 ± 0.008	0.73 ± 0.036	0.64 ± 0.012	0.26 ± 0.011	0.10 ± 0.006	0.22 ± 0.180
m=3	0.95 ± 0.004	0.98 ± 0.024	0.94 ± 0.018	0.53 ± 0.012	0.25 ± 0.007	0.42 ± 0.172
m=4	1.05 ± 0.008	1.06 ± 0.014	1.02 ± 0.022	0.72 ± 0.013	0.33 ± 0.009	0.49 ± 0.176
m=5	1.12 ± 0.004	1.19 ± 0.022	1.16 ± 0.020	0.87 ± 0.006	0.42 ± 0.007	0.51 ± 0.181
m=6	1.17 ± 0.006	1.22 ± 0.024	1.23 ± 0.025	1.04 ± 0.006	0.64 ± 0.017	0.52 ± 0.183

Similar to zero-sum games, Table 5 summarizes the Nash regrets found and compares it to our baselines. Since there is only one high-payoff yielding strategy in these games, random agents perform with much less variance in regrets, however, DQN agents are still unstable and suboptimal. Analogously, Figure 3 captures a similar summary to zero-sum games: in the top row we report independent co-training data, the bottom row contains fictitious self-play, while the left side contains Nash regrets and on the right average meta-episode reward is shown.

5.4 Discussion

Addressing **Q1**, we demonstrated that RL^2 shows promising convergence toward Nash equilibria in a wide variety of games, as evidenced by the decreasing Nash regrets, even in fully decentralized, independently trained agent settings.

However, the performance of RL^2 deteriorates rapidly as the complexity of the environment distribution increases, aligning with observations from prior work that identify the outer loop as a key bottleneck in meta-reinforcement learning. Despite this limitation, few approaches directly address this challenge [45], making it a compelling direction for future research. Nevertheless, algorithms with theoretical guarantees are still far to be surpassed by RL and meta-learning methods, especially as the action space size increases. Our results highlight both the potential and current limitations of RL^2 in multi-agent equilibrium learning, and reveal important insights about the co-adaptation of meta-trained RL agents.

Regarding **Q2**, we observe that RL^2 agents can quickly learn to exploit the underlying game structures not only in simple games, but complex families of games. However, this advantage is quickly fading as the action space size increases. This is partly due to the combinatorial complexity increase as the number of action grows, and partly due to the lack of any explicit communication between agents. Incorporating communication protocols [46, 47] into the meta-learning system might be useful to address this. Moreover, the decreasing speed of convergence as the number of actions increases also suggests that more complex models might be necessary for the meta-learners to be able to productively incorporate all information properly.

Addressing **Q3**, we find a clear and consistent advantage of independent co-meta-training over fictitious meta-self-play across both cooperative and competitive settings. The gap is particularly pronounced in cooperative games, where fictitious self-play performs significantly worse. This stems from the fact that discovering high-reward equilibria often requires simultaneous exploration by both agents, however this condition is not met when only one agent is learning while the other remains fixed. These findings suggest that the lack of mutual adaptation is a critical bottleneck in fictitious meta self-play. Nonetheless, increasing the

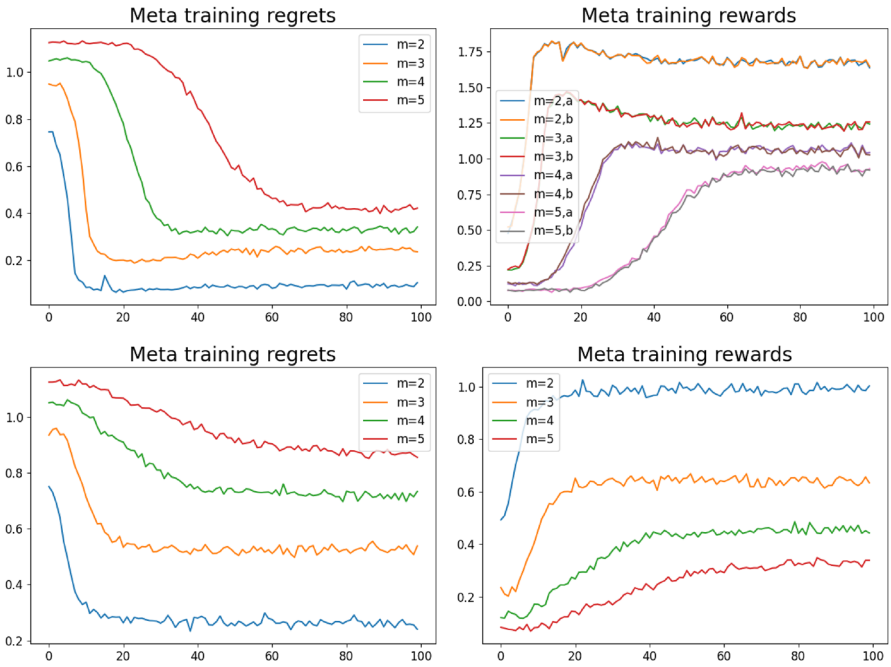


Fig. 3 The evolution of Nash regrets (left) and cumulative rewards (right) in cooperative games over outer-loop iterations (x-axis). Regrets/rewards are shown on the y axis

update frequency of the fixed agent may offer a promising path forward, potentially alleviating the asymmetry and fostering more stable incentives, whether to avoid being exploited in competitive games or to enable effective cooperation in collaborative ones.

6 Conclusions

In this work, we successfully adapted meta-learning methods to the multi-agent setting and proposed an interpretable evaluation scheme based on Nash regret and reward dynamics to assess their performance and gain insights into how they compete and cooperate.

To systematically study agent behavior in different strategic contexts, we introduced two types of families of games: zero-sum and noisy coordination games, capturing both competitive and cooperative dynamics. We compared independent, fully decentralized co-meta-training with fictitious meta-self-play, and further grounded our results with several baseline models, including random, value-based, recurrent agents and a theoretical baseline. Our findings highlight the advantages of co-meta-training in promoting generalizable equilibrium behavior, particularly in environments requiring joint adaptation. For future work, we envision exploring richer agent architectures, more expressive meta-learners, and adaptive opponent updating mechanisms to further improve stability and coordination. Indeed, many techniques and approaches from the literature seem promising in addressing the impediments encountered in our experiments. In particular, inter-agent communication protocols during meta-training could help address the lack of proper co-evolution in our fictitious self-play

experiments. Further, agent reputation protocols and the introduction of agent hierarchies might be useful in reaching more stable agent consensus in cooperative games.

Investigating multi-agent curricula and extending to sequential or partially observable settings could also broaden the applicability of meta-RL in decentralized scenarios.

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